Gurram Mohith Vamsi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.66
NARAYANA	Class XII (TSBIE)	2023	94.70%
Narayana co school lb,bairamalguda	Class X (SSC)	2021	10.00%

PROJECTS

Enterprise Document Intelligence Platform | Python · LangChain · ChromaDB · Groq · Docker

- Built a **RAG-based document question-answering system** for PDF ingestion and natural-language querying with **page-level source citations**.
- Implemented **Hybrid Search (BM25 + semantic vector search)** using **Reciprocal Rank Fusion (RRF)** and **Cross-Encoder reranking** to improve retrieval relevance.
- Developed **query rewriting and evaluation modules** to support multi-turn queries and track **hit rate, latency, and token usage**.

Mini Real-World Routing Engine | Java · OpenStreetMap

- Built a **shortest-path routing engine** on a real-world road network of **4,222 nodes and 8,080 directed edges** using Dijkstra's and A* algorithms.
- Implemented **Haversine-based A* heuristic** and a custom OSM parser supporting **weighted graphs and one-way roads**.
- Benchmarked both algorithms across **30 randomized queries**, reducing average nodes explored by **72.4%** with **zero correctness mismatches**, and visualized routes using Leaflet.js.

Spatio-Temporal Satellite Temperature Reconstruction | Python · XGBoost · Scikit-learn · MODIS

- Developed an **XGBoost spatio-temporal model** to reconstruct missing LST using spatial, temporal, and seasonal features.
- Engineered **15 spatio-temporal features** and implemented a **3-pass iterative gap-filling strategy** for different missing-data patterns.
- Achieved **R^2 = 0.925 on unseen 2021-2025 data** at 10% missingness, maintaining **R^2 = 0.871 at 30%**.

Evaluation of Forward Collision Warning Systems in Real-World Driving | ADAS · VBOX · Circuit Tools

*(**Supervised by** Dr. Digvijay Pawar)*

- Evaluated **driver response behavior to Forward Collision Warning (FCW)** systems across **34 drivers** under Stealth and Active warning phases.
- Analyzed driver responses across **urban arterial, expressway, and rural highway** conditions using VBOX and Circuit Tools data.
- Observed an increase in **Response With Warning from 7.0% to 10.4%** and a reduction in **No Response from 31.8% to 30.4%** during the Active Phase.

SKILLS

- **Programming Languages:** Java, Python, C
- **Web Technologies:** HTML, CSS, JavaScript
- **Database:** MySQL, SQL
- **Tools:** Git/GitHub, Jupyter Notebook, Microsoft Excel, Solid Edge, LaTeX
- **Data Analysis and Visualization:** NumPy, Pandas, Matplotlib, Power BI

RELEVANT COURSES

- **Programming courses:** Introduction to Programming, Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems.

- **Mathematics:** Calculus, Linear Algebra, Differential Equation, Probability, Statistics, Transform Techniques, Complex Variable.
- **Data Analytics & AI:** Introduction to Artificial Intelligence, Business Analytics with Python.

POSITIONS OF RESPONSIBILITY

- **Alumni cell,** Campus tour Coordinator : Organized and guided campus tours for visiting **alumni,** ensuring smooth coordination and engagement during events.
- **Prayas Club, Events Coordinator:** Managed and coordinated events focused on the education and development of children in nearby rural communities.
- **Otaku club,** Design Coordinator : Led creative design initiatives for club events and activities, managing posters, and promotional materials.

EXTRACURRICULAR

- **Volunteered 120+ hours** as part of the National Service Scheme (NSS), engaging in various community development activities.
- Participated in the **Clean India Drive,** promoting environmental awareness and cleanliness through hands-on involvement
- **Interests:** Pursue **volleyball** and **billiards** as recreational sports, reflecting an interest in physical activity, competition, and strategic thinking.